

section goes through the implementation details, and along the way you'll learn an interesting technique to allow for two-way communication between objects.

The scene manager is implemented in a class named `SceneMgr` in the `pyghelpers` module. As explained previously, in your main program, you create a single instance of the scene manager like this:

```
oSceneMgr = SceneMgr(scenesList, FRAMES_PER_SECOND)
```

The last line of your main program needs to be:

```
oSceneMgr.run()
```

[*Listing 15-6*](#) contains the code of the `__init__()` method of the `SceneMgr` class.

```
--- snip ---
def __init__(self, scenesList, fps):

    # Build a dictionary, each entry of which is a sceneKey : scene object
    ❶ self.scenesDict = {}
    ❷ for oScene in scenesList:
        key = oScene.getSceneKey()
        self.scenesDict[key] = oScene

        # The first element in the list is used as the starting scene
    ❸ self.oCurrentScene = scenesList[0]
        self.framesPerSecond = fps

        # Give each scene a reference back to the SceneMgr.
        # This allows any scene to do a goToScene, request, send,
        # or sendAll, which gets forwarded to the scene manager.
    ❹ for key, oScene in self.scenesDict.items():
        oScene._setRefToSceneMgr(self)
```

Listing 15-6: The `__init__()` method of the `SceneMgr` class

The `__init__()` method keeps track of all scenes in a dictionary ❶. It iterates through the list of scenes, asking each scene for its scene key, and